

JEE MAIN REVISION MASTER: SOME BASIC CONCEPTS OF CHEMISTRY 2026 2 QUESTIONS

3.33% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Some Basic Concepts Of Chemistry - Core Application	14 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Some Basic Concepts Of Chemistry - Core Application (High Yield — 14 PYQs)

Core Principle: Key problem pattern in Kcet Some Basic Concepts Of Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "From the given information, select the suitable law of chemical combination: Cupric Carbonate % of Cu % of C % of O Natural Sample 51.35 9.7..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following methods of expressing concentration are unitless?..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II and select the correct option (Based on mole concept): List-I List-II (a) 2 moles of ethene (i) 11.2 L volume at STP (b) Mass is equal to 66 g (ii) 56 g (c) 1 g of H_2 (iii) 12.04×10^{23} Molecules (d) 2 moles of water vapours (iv) 1.5 mole of CO_2

- (A) a-ii, b-iv, c-i, d-iii
- (B) a-iii, b-i, c-iv, d-ii
- (C) a-i, b-iv, c-ii, d-iii
- (D) a-ii, b-iii, c-i, d-iv

Correct Answer: None

Solution Analysis: For mole concept, we use these basic relations: 1 mole of any substance contains 6.02×10^{23} particles. 1 mole of any gas at STP occupies 22.4 L. Also, $\text{Mass} = \text{Number of moles} \times \text{Molar mass}$ For (a), ethene is C_2H_4 . Its molar mass is $2 \times 12 + 4 \times 1 = 28 \text{ g mol}^{-1}$ So, mass of 2 moles of ethene is $2 \times 28 = 56 \text{ g}$ Hence, (a) $\rightarrow$ (ii) For (b), mass is 66 g. This matches carbon dioxide, CO_2 , because molar mass of CO_2 is $12 + 2 \times 16 = 44 \text{ g mol}^{-1}$ Number of moles of